

Dams mark drought vulnerability rather than buffer it, and storage cannot offset a falling water supply in Chile

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Abstract

Expanding reservoir storage is the reflexive policy response to intensifying drought, yet whether dams build resilience or merely mark where vulnerability already concentrates remains untested under sustained drying. We treat reservoir presence as a basin-level intervention across central Chile (26 monitored reservoirs spanning the 2010–present megadrought) and estimate its effect on drought propagation and water-dependent land use using a national matched design (entropy balancing within Köppen $\times$ aridity strata; 21 of 24 treated basins retained, effective control sample ≈ 91), doubly-robust matched ATT, forcing-conditioned difference-in-differences, and randomization inference. Across every operational pathway (meteorological-to-hydrological drought transmission, evapotranspiration buffering, and irrigated-area expansion), the treatment effect brackets zero once siting is matched out. A within-basin placebo comparing regulated downstream reaches against physically unregulated upstream reaches of the same dammed basins (15 of 21 basins) is the cleanest within-design falsification: the apparent national buffering signal (slope gap -0.18) is as strong upstream as downstream and collapses to ≈ -0.03 within climate region ($p_{\text{perm}} = 0.39$), the signature of an aridity-gradient confound, not regulation. Refill efficiency is undiminished while the whole storage band drifts downward, a pattern consistent with declining water supply rather than a loss of storage or buffering capacity. Reservoirs in Chile thus appear to mark pre-existing, siting-determined vulnerability rather than make resilience; once siting is accounted for, the operational case for storage expansion as drought protection is not supported by these data.

Keywords: reservoir effect, drought propagation, socio-hydrology, causal inference, megadrought, Chile

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1. Introduction

Reservoirs are the dominant structural response to drought: when water shortages inflict economic damage, public pressure tends to drive the expansion of storage to raise water availability, which in the short term lowers the frequency, severity and duration of shortage (Di Baldassarre et al., 2018). Yet a socio-hydrological perspective warns that the same infrastructure can be self-defeating over the planning horizon of a dam (20–30 years) through two coupled feedbacks. In the *supply–demand cycle*, greater supply enables agricultural, urban or industrial expansion, so demand grows faster than projected from socio-economic trends alone and quickly offsets the initial benefit (a rebound, or Jevons, effect) (Di Baldassarre et al., 2018). In the *reservoir effect*, the drought analogue of the levee effect, prolonged reliance on abundant stored water erodes the incentive to adapt, deepening dependence on infrastructure and amplifying the damage when a drought finally exceeds storage capacity (Di Baldassarre et al., 2018). Global accounting is consistent with this view: water demand has outpaced storage capacity in recent decades, and local water histories from Las Vegas, Melbourne and Athens show consumption rising to meet, and then exceed, each new increment of supply (Di Baldassarre et al., 2018), and the storage returns of new dam construction are themselves diminishing worldwide: reservoirs built since 1999 hold a markedly smaller fraction of their capacity than older ones (mean normalized storage 61 % versus 70 %), and South America records among the steepest declines on record (a drop of 4.3 % over two decades) (Li et al., 2023). Critically, the rebound means the enhanced damage from over-reliance need not be offset by a lower frequency of shortage, making the safe-development paradox potentially *more* dangerous for drought than for floods (Di Baldassarre et al., 2018). Whether these feedbacks dominate in practice, or are outweighed by reservoirs’ protective function, remains actively debated even as much of the existing dam stock ages past its design life and the case for further storage is openly questioned (Ho et al., 2017; Kellner, 2021).

The short-term, hydrological side of this ledger is well documented. As drought propagates from precipitation deficit through soil moisture to streamflow, modulated by catchment characteristics (Van Loon, 2015; Van Loon and Laaha, 2015; Wang et al., 2016), comparisons of regulated against naturalized streamflow show that reservoir operation reduces the duration and magnitude of short-term (1–3 month) hydrological drought downstream and lengthens the lag between meteorological and hydrological drought (Wu et al., 2017, 2018), and that reservoir management can damp rapid drought-to-flood transitions, particularly in snowmelt-driven catchments (Götze and Brunner, 2024). But the same regulation can redistribute scarcity in space and time (Van Loon and Van Lanen, 2013): in the transboundary Tagus, operation of the Alcántara dam lengthened

and intensified droughts *downstream* of the reservoir relative to the pre-dam period (López-Moreno et al., 2009). The long-term, demand side is far less constrained empirically. Irrigation modernization that nominally “saves” water often expands irrigated area or shifts farmers toward higher-value, more water-intensive crops (the efficiency paradox), raising consumptive use and basin-scale vulnerability rather than lowering it (Grafton et al., 2018; Jordán et al., 2025). Whether reservoirs set the same trap remains largely untested.

Chile is an acute testbed. Since 2010 central Chile (30–38° S) has endured a *megadrought*: an uninterrupted run of dry years with mean rainfall deficits of 25–45 %, punctuated by “hyperdrought” winters in 2019 and 2021 with deficits exceeding 75 % (Garreaud et al., 2025, 2017). The event is unprecedented over the instrumental record and has no clear analogue in millennial tree-ring reconstructions; roughly one-quarter to one-third of its intensity is attributable to anthropogenic forcing (in line with warming-driven increases in drought risk documented globally (Diffenbaugh et al., 2015; Trenberth et al., 2014)), the remainder to Pacific decadal variability and long-term regional drying (Boisier et al., 2018; Garreaud et al., 2017; Montecinos and Aceituno, 2003; Quintana and Aceituno, 2012). The signal amplifies as it propagates: a ~30 % reduction in Andean snow-water equivalent (Saavedra et al., 2018), part of intensifying snow drought worldwide (Cowherd et al., 2023; Huning and AghaKouchak, 2020), translates into streamflow declines of 20–70 % (locally up to 90 %), with the streamflow deficit in arid sectors often twice the rainfall deficit, and reservoirs and groundwater drawn to historical lows (Alvarez-Garreton et al., 2021; Garreaud et al., 2017), part of a global pattern in which 93 % of monitored reservoirs have failed to refill at least once and the few showing sustained level declines cluster in the 14–44° S band that encloses central Chile (Wang et al., 2024). Demand, however, has scarcely relented. Infrastructure and intensive groundwater pumping have held human withdrawals near pre-drought levels, masking a “Day Zero” that surface water alone could not avert (Alvarez-Garreton et al., 2021; Garreaud et al., 2025), at the cost of aquifer overexploitation now well documented in central Chile (Duran-Llacer et al., 2020; Taucare et al., 2024; Venegas-Quiñones et al., 2024). Mounting evidence suggests the region is crossing from episodic drought into structural aridification, with long-term declines in water supply (and rising atmospheric evaporative demand) driving land-cover change (Fuentes et al., 2021; Running, 2022; Vicente-Serrano et al., 2022; Zambrano et al., 2025).

These dynamics are inseparable from Chile’s water governance. The 1981 Water Code, the paradigm case of market-based water management, made water-use rights tradable, perpetual private property separable from land (Macpherson et al., 2023; Reade Malagüeño and D’Odorico, 2024). Rights were granted on the stationarity assumption that historical flows would persist, producing over-allocated “paper rights” in many

basins, while management was delegated largely to private water-user organizations (Barría et al., 2021; Rivera et al., 2016; Rivera-Vidal et al., 2025; Valdés-Pineda et al., 2014). Because the state cannot readily curtail existing private rights during scarcity, the policy reflex has been supply-side: a national programme of new dams and inter-basin transfers (Barría et al., 2021; Garreaud et al., 2025). The same institutional setting rewarded conversion to high-value, water-demanding export crops (orchards and vineyards), concentrating water in agro-export firms and locking in consumptive demand even as supply contracted (Reade Malagueño and D’Odorico, 2024; Webb et al., 2021). Yet most farmers still meet scarcity with short-term, reactive measures rather than durable adaptation, leaving long-run vulnerability largely intact (Jordán et al., 2025). Chile thus exhibits, in compressed form, every ingredient of the reservoir effect: an extreme and persistent forcing, a supply-side infrastructure response, market incentives for demand expansion, and weak adaptive feedback.

Despite its prominence, the reservoir effect for drought has been described far more than it has been *measured*. The framework’s authors note that we lack the datasets and analytical tools to quantify these social–technical–hydrological feedbacks, and, above all, defensible counterfactuals: what a drought’s impact *would have been* in a basin had the reservoir not been built (Di Baldassarre et al., 2018). Existing work compounds the difficulty. Drought- propagation studies remain mostly temporal, leaving the spatial footprint of upstream regulation on downstream drought poorly resolved; where it has been traced, severe upstream droughts can propagate downstream with added severity and duration rather than being buffered away (Schilstra et al., 2024). Quantitative comparisons of drought resistance across dams are likewise scarce, and the few that exist show that buffering is neither uniform nor guaranteed: it weakens or even reverses as drought intensifies, so that the very protective function invoked to justify storage is least reliable in the extreme events that matter most (Lee et al., 2026). The core obstacle is identification: reservoirs are not sited at random but built where irrigation demand and aridity already concentrate, so dammed basins differ systematically from undammed ones before any drought strikes. Basin-mean greenness, raw before–after contrasts, or calendar-time comparisons cannot separate a genuine reservoir effect from this siting confound.

Here we ask whether reservoirs build resilience or merely postpone and concentrate vulnerability during prolonged drought. We treat the presence of a major reservoir as a basin-level intervention and construct the missing counterfactual by pairing a matched dammed-versus-undammed design with a forcing-conditioned estimand, comparing how dammed and hydroclimatically comparable undammed basins respond to the *same* drought forcing. This design targets the long-term demand-side feedback that prior, hydrology-only analyses

leave open, and turns the socio-hydrological reservoir-effect hypothesis into a falsifiable, causally framed test in the setting where it matters most.

2. Methods

2.1. Study area and design

We study continental Chile (18–44° S), a strong north–south aridity gradient spanning hyper-arid, Mediterranean, and humid-temperate climates, over 2005–2024, the window in which reservoir storage, meteorological forcing, and satellite outcomes jointly exist (`config/study_period.yml`). The design is a matched comparison of **dammed versus undammed sub-watersheds**, with reservoir presence treated as a basin-level intervention. Because 18 of 24 monitored reservoirs were commissioned before the storage record begins, treatment is effectively time-invariant over the window and a staggered-adoption (commissioning) event study is infeasible; identification therefore rests on a matched cross-section combined with conditioning on meteorological forcing rather than on calendar time.

The central estimand is the operational effect of a reservoir on drought outcomes *conditional on the covariates that govern where reservoirs are sited*, that is, whether dammed basins respond to the same drought forcing differently from hydroclimatically comparable undammed basins. This isolates reservoir operation from reservoir siting (Di Baldassarre et al., 2018).

2.2. Spatial units and treatment assignment

The analysis grain is the DGA BNA **sub-watershed (subcuenca)** (467 units), chosen over the coarser cuenca grain (101 units) for power and common support: reservoirs sit in Chile’s largest main-stem basins, and the finer grain supplies far more controls inside the treated size range (`docs/design/grain-selection.md`). A subcuenca is **treated** if it contains 1 of the 26 monitored reservoirs by point-in-polygon assignment, giving **24 treated units**. An untreated subcuenca that lies within a cuenca containing a reservoir is flagged as **contaminated** (it shares the regulated main stem and is not a clean counterfactual) and removed (45 units), leaving **398 clean controls**. This contamination rule defuses the most direct downstream-spillover (SUTVA) violation; residual adjacency and shared-aquifer spillovers between neighbouring subcuencas are carried as an identification caveat and checked at cuenca grain.

2.3. Data

Reservoir storage and treatment. Monthly storage for 26 monitored reservoirs (2005-01 to 2026-04) with a per-reservoir capacity column (`max_level_hm3`); storage and capacity share units (hm³), so percent-of-

capacity ($100 \times \text{level} / \text{max_level_hm3}$) is the cross-reservoir-comparable state variable. Static attributes (commissioning year, use, size class) come from the national dam registry.

Meteorological forcing. Standardized drought indices (SPI, SPEI, and EDDI) derived from CHIRPS precipitation and CHIRTS temperature at timescales of 1/3/6/12/24/36 months, standardized on the 1991–2026 baseline (`config/variables.yml`). SPEI (Vicente-Serrano et al., 2010) at 12 months is the primary exogenous dose; the suitability of CHIRPS-derived standardized indices for Chilean drought monitoring is established, including ENSO-modulated drought variability across the country (Meza, 2013; Oertel et al., 2020; Zambrano et al., 2017). We use the 12-month accumulation because it matches the annual, snowmelt-fed hydrological cycle of central Chile (winter precipitation stored as snow and released through the growing season) and so captures the inter-seasonal carryover that irrigation reservoirs are built to bridge, at the scale at which multi-year vulnerability (rather than sub-seasonal operation) is expressed. By construction this does not resolve short-term (1–3 month) operational buffering, which reservoirs are documented to provide (Wu et al., 2017, 2018) and which we flag as a boundary of our conclusions (Discussion). Conditioning on forcing (estimating effects on the deficit-to-impact *slope* rather than on levels) is how we break the collinearity between the 2010-onset megadrought and the treatment clock.

Hydrological drought. Daily streamflow from the CR2 network (2000–2020) is aggregated to the Standardized Streamflow Index at 12 months [SSI-12; Vicente-Serrano et al. (2012); Lema et al. (2025)], the regulated-flow analogue of SPEI-12, used as the outcome in the within-basin placebo (below). Gauges are classified as upstream or downstream of each dam from an SRTM digital elevation model.

Ecological and land-use outcomes. (i) annual irrigated/cropland area from MapBiomass Chile Collection 2 (30 m, 1999–2024); (ii) actual evapotranspiration from MODIS (8-day, ~500 m, 2001–2024), for whole basins and for an orchard stratum from the CIREN/ODEPA cadastre; (iii) standardized vegetation and productivity anomalies, namely monthly z-score accumulated NDVI (zcNDVI-6), annual z-score NPP (zNPP), and the standardized actual-ET anomaly (SETI), from MODIS over 2000–2024; the zcNDVI productivity proxy is validated for Chilean agricultural drought, and the ecological response of Chilean vegetation to the megadrought is well characterized (Glade et al., 2016; Venegas-González et al., 2023; Zambrano et al., 2018).

Matching covariates. Baseline aridity (long-term mean annual P/PET over the 1991–2020 WMO normal), watershed area, mean elevation (SRTM), and Köppen–Geiger climate class, extracted per subcuenca by zonal statistics.

2.4. Matched control construction

We estimate the average treatment effect on the treated (ATT) using **entropy balancing** (`WeightIt`, `method = "ebal"`) exact-matched within Köppen main group, balancing **log(area)** and **mean elevation** (`docs/design/matched-controls.md`). Balancing on size and elevation rather than latitude is deliberate: latitude and the climate class fight each other across the Andes–Patagonia span, and substituting SRTM elevation raised the effective control sample size from 9 to 91 while balancing latitude for free.

The procedure retains **21 of 24 treated subcuencas** (three high-Andes units fall outside local elevation common support) and achieves essentially perfect balance (standardized mean differences for `log(area)` and elevation fall from 1.18 / 0.37 to 0.000, variance ratios 1), with an **effective control sample size 91** out of 244 in-window controls. Baseline aridity is a genuine confound (dammed basins are markedly more arid: median P/PET 0.26 vs 0.80) but is *not* used as a balance target, because hard-balancing it collapses the ESS from 91 to 19; instead the residual aridity imbalance (SMD 0.17, above the 0.1 target) is absorbed by doubly-robust regression adjustment. Because the matched set does not fully balance aridity, this adjustment is *load-bearing* rather than cosmetic, and the estimates can be sensitive to its functional form; we therefore stress-test the aridity-adjusted estimates with a quadratic aridity term and report the curvature-robust values where feasible (see Results). Robustness to the matching method is assessed with coarsened exact matching (CEM) and 1:2 nearest-neighbour (Mahalanobis) matching on the identical common-support sample; entropy balancing and NN agree on all 21 treated, while CEM additionally prunes the 4 most extreme large/high basins, so all treatment effects are reported with and without those units.

2.5. Causal estimands and estimators

Doubly-robust matched ATT. Each outcome is estimated three ways on the same sample, comprising weighting-only (ebal-weighted difference in means), regression-only (covariate-adjusted g-computation), and the headline **doubly-robust** estimator (ebal weights *and* an outcome model in `log(area)`, elevation, and `log(aridity)`), so functional-form sensitivity is visible. Continuous covariates are centred at the treated mean so the treatment coefficient reads as the ATT; standard errors are HC3-robust (ebal weights treated as fixed, the standard mildly anti-conservative simplification).

Forcing-conditioned transmission slope. To remove megadrought *exposure* (concentrated in exactly the arid basins where reservoirs sit), the outcome for the DR machinery is the per-basin **transmission slope**, the regression of an annual outcome (e.g. zNPP) on annual SPEI-12 over 2005–2024, a standardized,

cross-basin drought-transmission elasticity. The vulnerability hypothesis predicts a *steeper* slope (more impact per unit deficit) in dammed basins.

Forcing-interacted difference-in-differences. On the observed area and ET panels we fit

$$y_{it} = \beta (\text{dammed}_i \times \text{SPEI}_{it}) + \gamma \text{SPEI}_{it} + \alpha_i + \tau_t + \varepsilon_{it},$$

with subcuenca (α_i) and year (τ_t) fixed effects and entropy-balancing weights. Because treatment is time-invariant, the dose is meteorological SPEI rather than calendar year: the year fixed effects absorb the common megadrought shock, and β is the differential deficit-to-impact slope between dammed and matched-control basins, robust to the siting-in-arid-central-Chile confound, which biases levels far more than slopes. The vulnerability hypothesis predicts $\beta > 0$.

Equivalence testing and minimum detectable effect. Because the headline results are nulls, we move beyond failure-to-reject and bound the operational effect. For each null we report a two one-sided test (TOST): the effect is declared equivalent to zero if its 90% confidence interval lies within a negligible region of $\pm 25\%$ of the baseline (untreated) transmission slope. We are not aware of an established hydrological or operational standard defining a “negligible” reservoir effect on drought transmission, so we anchor the margin in the granularity of operational drought classification rather than choosing it arbitrarily. Standardized drought indices are managed in discrete severity categories each 0.5 standardized units wide (McKee et al., 1993; Vicente-Serrano et al., 2010); over the observed range of meteorological deficits a change of $< 25\%$ in the deficit-to-impact slope (0.13 SSI units on the 0.53 baseline) alters the propagated streamflow-drought index by less than a single severity category, i.e. it cannot move a basin across an operational drought-severity threshold and is therefore marginal for drought management. We adopt $\pm 25\%$ as this deliberately strict, pre-specified margin and, because the equivalence verdict is nonetheless sensitive to the bound, also report the threshold-free MDE so that readers may apply any margin they consider material. The qualitative conclusion (only large effects are excluded; no outcome is equivalent at $\pm 25\%$) does not depend on the exact margin. The minimum detectable effect (at 80% power, two-sided $\alpha = 0.05$) and the equivalence interval are built from the **standard deviation of the permutation-null distribution**, not the analytic standard error, since cluster-robust SEs over-reject at ~ 21 clusters (`src/R/causal/equivalence.R`). To confirm the null is informative rather than underpowered, we additionally run a **positive control**: a known buffering slope is injected into the treated units and the full estimator and randomization inference are re-run, verifying

that the design recovers an effect of policy-relevant size and correctly flags none when none is injected.

Siting-confound decomposition. To quantify how much of any apparent buffering is selection, we fit a ladder on the same panel: (1) a naive pooled slope gap (no weights, no fixed effects, what a conventional dammed-vs-undammed study reports); (2) unit + time fixed effects, unweighted; (3) the design estimator (ebal weights + fixed effects). The shrinkage from (1) to (3) is the siting confound; the residual at (3) is the regulation estimate.

Within-basin upstream/downstream placebo. The primary selection-versus-effect discriminator contrasts the SPEI-12→SSI-12 transmission slope at regulated **downstream** reaches against physically unregulated **upstream** reaches of the *same* dammed basins, which share climate, geology, and selection history. A genuine regulation effect implies a downstream-specific dose-response that the upstream placebo does not reproduce; an attenuation equally present upstream is siting, not regulation. Of the 21 treated matched basins, 15 have both an upstream and a downstream gauge, giving near-national coverage rather than a single case study. The contrast is also split by Köppen region to test whether a regulation effect emerges in the high-supply south. Because upstream gauges sit at higher elevations than downstream gauges, we test the placebo’s comparability assumption directly: we quantify the elevation gap and regress the per-gauge transmission slope on elevation among undammed control gauges, so that any elevation-driven regime difference is measured rather than assumed.

2.6. Inference

With only ~21 treated clusters, cluster-robust standard errors over-reject, so the primary inference is **randomization inference**: the treatment label is permuted among units within Köppen × aridity- tercile strata (999 permutations) and the observed statistic is compared against the resulting null distribution; cluster-robust confidence intervals are reported alongside for completeness. Parallel pre-trends are checked with a dynamic event study (dammed × year, reference 2009), and residual confounding is probed with an aridity placebo that relabels the most-arid control tercile as pseudo-treated.

2.7. Software and reproducibility

All analyses are implemented in R within a **targets** pipeline (matched-set construction, estimators, and figures are individual targets), using **WeightIt/cobalt** for balancing and diagnostics, **fixest** for fixed-effects regression, **sf/terra/exactextractr** for the spatial extractions, and **renv** for package pinning. Every result in the manuscript corresponds to a named pipeline target; no figure or table is produced by hand.

3. Results

3.1. Reservoirs mark, but do not make, irrigated vulnerability

Across the matched design (21 dammed vs 244 control subcuencas, effective control sample 91 after entropy balancing), dammed basins contain roughly 4.6 times more cropland than matched controls, a mean of 11.1% of basin area versus 2.4% (Figure 1, panel a). This gap, however, is a *siting* signature: it is present and stable across the entire 2000–2024 record, with dammed and control trajectories evolving in parallel through the 2010-onset megadrought (dashed line). A dynamic event study confirms this. The dammed $\times$ year coefficients on cropland area are jointly flat before 2009 (Wald $p = 0.68$) and show no divergence after the drought onset (Figure 1, panel b). Reservoirs mark where irrigated agriculture already was; they do not detectably expand it or change how it responds to drought.

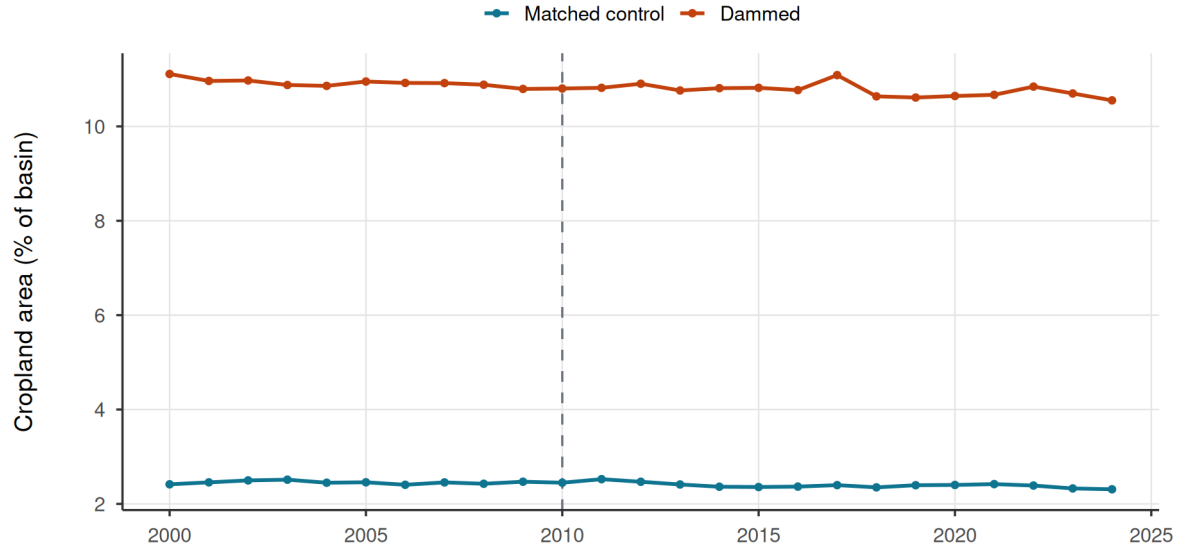
3.2. No operational reservoir effect survives once siting is matched out

We probe the operational vulnerability mechanism with four independent estimators (Table 1): two cross-sectional doubly-robust matched ATTs, namely net irrigated-area expansion (-0.69 ha km $^{-2}$, 95% CI $[-2.4, 1.1]$) and reservoir evapotranspiration buffering (-0.014 d log ET / d SPEI, $[-0.065, 0.037]$), and forcing-interacted difference-in-differences slope gaps on the observed area and ET panels, in which the deficit dose is the meteorological SPEI rather than calendar time (so year fixed effects absorb the common megadrought shock and the estimand is the differential deficit-to-impact *slope*). Because the treatment is effectively time-invariant (18 of 24 reservoirs predate the record), this within-unit, forcing-based design is the only defensible dynamic identification; inference uses randomization of treatment within climate $\times$ aridity strata (~ 21 treated clusters).

Under the design’s primary inference, no estimate indicates an operational effect (Figure 2, Table 1). Three of the four straddle zero outright: net irrigated-area expansion, and the slope gaps for irrigated area ($p_{\text{perm}} = 0.91$) and orchard ET ($p_{\text{perm}} = 0.24$). The whole-basin ET slope gap is the exception, and we treat it explicitly rather than fold it silently into the nulls. Its cluster-robust interval is positive and excludes zero ($+0.048$, $[0.026, 0.070]$); however, it fails the design’s pre-specified randomization inference ($p_{\text{perm}} = 0.11$) and, decisively, its dynamic event study shows strongly non-parallel pre-trends ($p \approx 10^{-10}$). The panel is therefore confounded (whole-basin MODIS ET is contaminated by residual aridity over sparsely vegetated, barren land), so neither its point estimate nor its cluster-robust interval is interpretable, and for the same reason it is the one outcome we exclude from the equivalence bounding below. Two facts keep this exception from rescuing a buffering narrative: taken at face value the estimate points toward *vulnerability* (a

Reservoirs do not alter irrigated-area dynamics

a



b

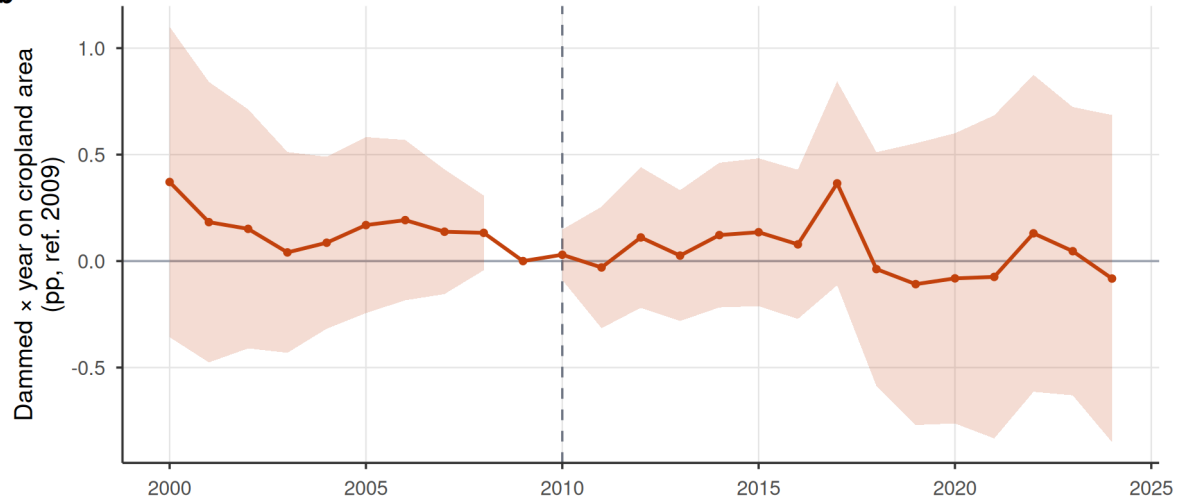


Figure 1: Observed irrigated-area dynamics in dammed versus matched-control basins. (a) Weighted mean cropland area (% of basin, MapBiomass class 18); dammed basins hold ~4.6x more cropland (11.1% vs 2.4% of basin area) but trajectories are parallel. (b) Dynamic event study (dammed x year, reference 2009): flat pre-trends and no post-megadrought divergence; shaded band is the 95% confidence interval (coefficient ± 1.96 SE). Dashed line, 2010 megadrought onset.

steeper deficit-to-impact slope), not buffering; and the cleaner orchard-ET stratum and the cross-sectional ET-buffering ATT both straddle zero. Across the interpretable outcomes, then, no panel shows the positive slope the vulnerability mechanism predicts nor the negative slope operational buffering would produce: conditional on siting, the operational reservoir effect brackets zero.

Table 1: Reservoir-effect estimates across estimators. Cross-sectional doubly-robust matched ATTs and forcing-interacted DiD slope gaps (treat:SPEI). DiD rows are judged by randomization-inference p-values (p), the design’s primary inference at ~21 clusters; cluster-robust CIs are shown for completeness and over-reject here. A positive estimate would indicate induced-demand vulnerability; a negative one, operational buffering. *Whole-basin ET fails its event-study pre-trend test (p approximately 1e-10): its cluster-robust CI excludes zero but the panel is confounded by barren-land aridity, so it is labelled confounded and excluded from the equivalence bounds (Table 2).

Group	Outcome	Estimand	Esti-		p	Verdict
			mate	95% CI		
Cross-sectional matched ATT	Irrigated-area expansion	ha km ⁻² added	- 0.687	[-2.44, 1.06]	-	null
Cross-sectional matched ATT	Reservoir ET buffering	d log ET / d SPEI	- 0.014	[-0.0646, 0.0369]	-	null
Forcing-interacted DiD (slope-gap)	Irrigated area	differential deficit->impact slope	0.001	[-0.00344, 0.00493]	0.91	null
Forcing-interacted DiD (slope-gap)	Whole-basin ET	differential deficit->impact slope	0.048	[0.0255, 0.0703]	0.11	con- founded*
Forcing-interacted DiD (slope-gap)	Orchard ET	differential deficit->impact slope	- 0.020	[-0.0448, 0.00565]	0.24	null

3.3. Sensitivity to aridity adjustment

Dammed basins are systematically more arid than controls, and the matched set does not fully remove this: baseline aridity remains imbalanced after weighting (standardized mean difference 0.17, above the 0.1 target), so the doubly-robust adjustment for aridity is load-bearing rather than a cosmetic mop-up. Because dammed basins are the more arid, residual under-control biases *toward* apparent vulnerability, which makes the nulls conservative. Functional form matters and we report it: in the ecological-proxy transmission-slope analysis the dammed-versus- control estimate falls from +0.33 under linear aridity adjustment to +0.11 once a quadratic aridity term is added, and in the irrigation-specific stratum it collapses to 0 (0.08 → 0.01). The apparent steepening is therefore largely a water-limitation/aridity artifact (consistent with the siting

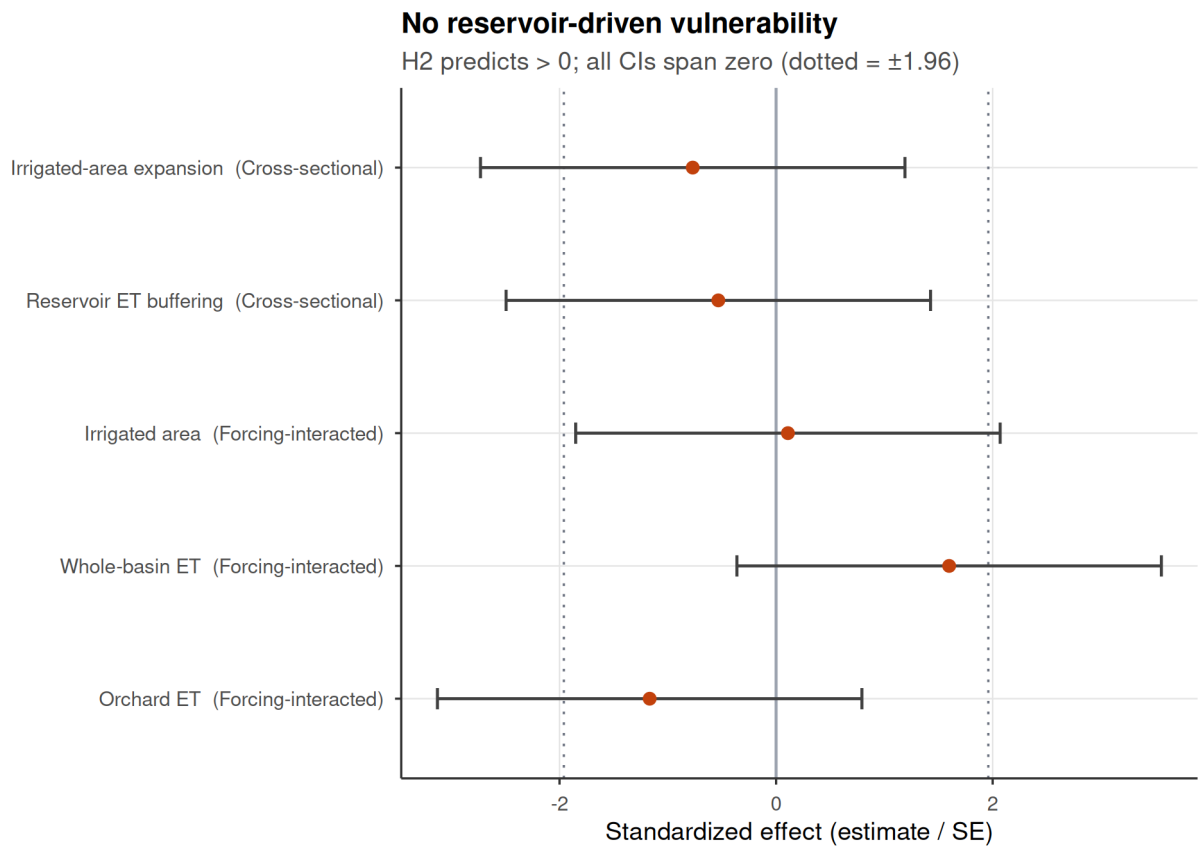


Figure 2: Standardized dammed-versus-control effect across the four estimators. DiD rows are standardized via their randomization-inference p-value (the design's primary inference), so whole-basin ET (cluster-significant but permutation-null and pre-trend-confounded) sits within the bounds; cross-sectional ATTs use estimate/SE. Horizontal bars span ± 1.96 standardized units (the 95% reference interval); dotted lines mark ± 1.96 .

interpretation), but the same sensitivity means estimates that rely on linear aridity control should be read as functional-form-dependent, and we treat the curvature-robust values as the conservative reference.

3.4. Bounding the null: what the design can rule out

A null is only informative against a stated effect size, so we convert each non-result into a bound via equivalence testing (two one-sided tests) and a minimum detectable effect (MDE) built from the permutation-null spread (Table 2). We bound the four interpretable outcomes (streamflow ITT and downstream, irrigated-area DiD, orchard-ET DiD); whole-basin ET is excluded because its pre-trends fail, so an equivalence interval on a confounded estimate would be meaningless. The streamflow design has the most power: it can detect a buffering effect that attenuates SPEI→SSI transmission by 46% of the baseline slope (ITT MDE 0.27 against a baseline transmission of 0.59; downstream-only MDE 0.31 against its 0.53 baseline, 58%). Here “baseline” is the untreated (control) SPEI→SSI slope from the fixed-effects model (Table 2), which is distinct from, and should not be confused with, the raw per-gauge mean transmission of downstream gauges (0.599) reported for the placebo comparability check (Table 4). No outcome’s 90% interval falls inside its pre-specified negligible region ($\pm 25\%$ of baseline), so we do *not* claim formal statistical equivalence to zero (with ~21 treated clusters the design can exclude only **large** operational effects, not small ones). A policy-relevant buffer of 20–45% of baseline transmission is *not* excluded. The causal weight of the null therefore rests on the within-basin placebo below, not on equivalence: the design excludes only large operational effects, and the placebo indicates that the residual apparent attenuation is selection rather than regulation.

Table 2: Equivalence / minimum-detectable-effect bounds on the headline nulls. Observed slope gap, MDE (smallest effect detectable at 80% power from the permutation-null SD), the 90% (TOST) interval, and whether it lies within the negligible region ($\pm 25\%$ of baseline). ‘baseline’ is the untreated (control) SPEI→SSI / SPEI→impact transmission slope from the fixed-effects model. No outcome is formally equivalent to zero; the streamflow design rules out attenuations larger than ~46-58% of baseline transmission.

Outcome	Observed	MDE	90% CI	Equiv. to 0?	p_perm
streamflow SSI (ITT)	-0.183	0.268	[-0.34, -0.0253]	no	0.39
streamflow SSI (downstream)	-0.165	0.312	[-0.348, 0.0186]	no	0.48
irrigated area (DiD)	0.001	0.010	[-0.00506, 0.00656]	no	0.90
orchard ET (DiD)	-0.020	0.046	[-0.0463, 0.00718]	no	0.25

To confirm that this null is informative rather than merely underpowered (that the design could detect a true effect of the relevant order), we ran a positive control on the streamflow estimator (Table 3). Injecting a known

buffering slope into the treated downstream units, the estimator recovers it almost exactly (a -0.30 injection yields -0.47 on top of the -0.17 baseline), the no-injection case is correctly *not* flagged ($p_{\text{perm}} = 0.48$), and randomization inference detects the effect once the total slope reaches -0.47 ($p_{\text{perm}} = 0.014$), with -0.62 strongly detected ($p_{\text{perm}} = 0.001$). The design therefore has the power to reveal a large operational buffering effect where one exists, so its failure to find one is evidence of absence at that magnitude (consistent with the MDE above) rather than blindness. This positive control validates the streamflow estimator specifically; the irrigated-area and ET nulls are not independently power-validated, their minimum detectable effects being large (for irrigated area, far above any plausible effect; Table 2), and rest on the placebo and the conservative direction of the aridity bias rather than on their own power.

Table 3: Positive control on the streamflow estimator. A known buffering slope (beta_injected; negative = buffering) is added to treated downstream units; the estimator recovers it (recovered = baseline $-0.17 + \text{beta}$), the no-effect case is not flagged, and randomization inference detects the effect once it exceeds the minimum detectable effect.

beta_injected	recovered	p_perm	detected
0.00	-0.165	0.480	FALSE
-0.15	-0.315	0.135	FALSE
-0.30	-0.465	0.014	TRUE
-0.45	-0.615	0.001	TRUE

3.5. The apparent buffering is reservoir siting, not regulation

The preceding nulls use vegetation and ET proxies. We obtain the same answer on the direct, vegetation-independent availability outcome (streamflow), and, uniquely here, can show *why* a naive analysis would mislead. From CR2 daily gauge records (2000–2020) we build a 12-month Standardized Streamflow Index (SSI-12) and test whether meteorological drought (SPEI-12) propagates less strongly to streamflow drought in dammed than in matched-control basins (Figure 3). Dammed basins do show a flatter SPEI→SSI transmission, the visual signature of buffering (Figure 3, panel a). But this apparent attenuation **(i)** does not survive randomization inference (intent-to-treat slope gap -0.18 , cluster $p = 0.03$ but $p_{\text{perm}} = 0.39$), and **(ii)** is present just as strongly at gauges located *upstream* of the dam (which are physically unregulated) as at downstream gauges (-0.20 vs -0.16 ; Figure 3, panel b). There is no downstream-versus-upstream dose-response, the signature a genuine regulation effect would require. The flatter slope is therefore a property of *where* reservoirs are sited (arid, hydrogeologically distinct basins), not of the reservoirs themselves.

340 The siting interpretation is confirmed by the way the signal behaves across the aridity gradient. The
341 national “buffering” slope gap (−0.16 to −0.20) **collapses to −0.03 once the comparison is made**
342 **within climate region**, the textbook signature of a between-region confound, in which arid treated basins
343 are contrasted against a control pool spanning the whole wet→dry gradient. In the high-supply south, where
344 buffering should be easiest to detect, the downstream and upstream gaps are statistically identical (−0.036 vs
345 −0.034) and both indistinguishable from zero (southern downstream $p_{\text{perm}} = 0.78$). Splitting by region thus
346 both kills the signal and equalizes the upstream and downstream reaches, two independent confirmations
347 that the attenuation is the aridity gradient, not regulation. Because the upstream-of-dam comparison is
348 unavailable for any vegetation outcome, streamflow provides the cleanest within-design falsification: the
349 effect a naive dammed-versus-control or calendar-time analysis would report is siting, not regulation.

350 The placebo assumes upstream and downstream reaches are hydrologically comparable, which we test
351 directly (Table 4). Upstream gauges do sit higher than downstream ones (mean 1.0 vs 0.46 km), and gauge
352 elevation is not innocuous. Among undammed control gauges the SPEI→SSI transmission slope declines by
353 0.21 per kilometre of elevation ($p < 0.001$). The placebo nonetheless holds, for two reasons. First, the *raw*
354 transmission slopes of upstream and downstream treated gauges are nearly identical (0.565 vs 0.599), so the
355 elevation difference does not produce a transmission-regime gap of the size at issue. Second, the apparent
356 buffering is if anything *stronger* upstream (−0.20) than downstream (−0.16), the opposite of what regulation,
357 which acts only downstream, would produce. We cannot claim the reaches are perfectly interchangeable, and
358 a residual elevation/regime component remains a caveat, but neither the raw slopes nor the direction of the
359 gap supports a regulation effect masked by elevation.

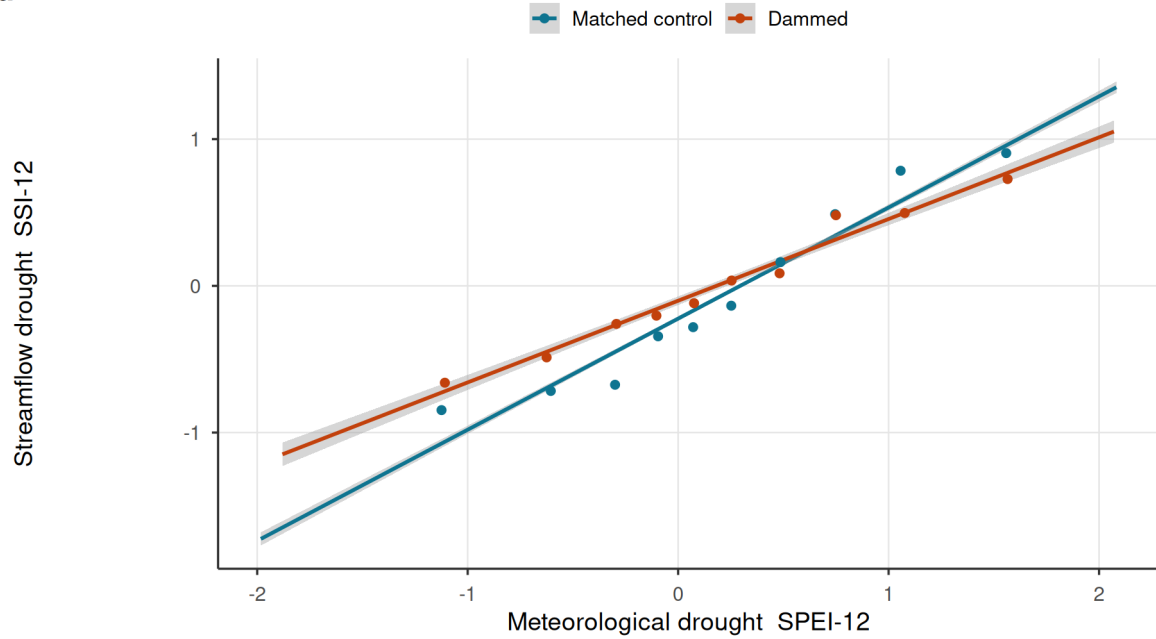
Table 4: Upstream/downstream placebo comparability. Upstream gauges are higher than downstream, and gauge elevation predicts the transmission slope among undammed controls (−0.21 per +1000 m), yet upstream and downstream treated gauges have nearly identical raw transmission slopes, so the equal up/down attenuation is not an elevation artifact.

quantity	value	detail
upstream gauges (elev, mean m)	1003.000	n=28; mean transmission 0.565
downstream gauges (elev, mean m)	459.000	n=40; mean transmission 0.599
control transmission slope vs elevation (per +1000 m)	−0.212	p = 0.00 (n=84 control gauges); ~0 => elevation does not drive transmission

Apparent streamflow buffering is siting, not the reservoir

Effect is permutation-null and as strong in UPSTREAM (unregulated) gauges

a



b

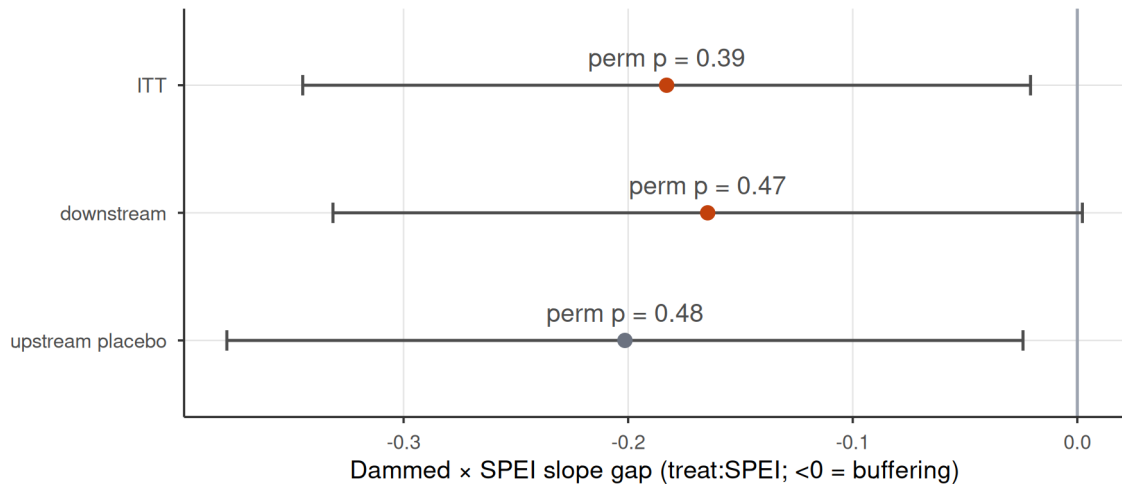


Figure 3: Streamflow buffering is reservoir siting, not regulation. (a) SPEI-12 to SSI-12 transmission for dammed versus matched-control basins (downstream gauges); lines are weighted least-squares fits and shaded bands are 95% confidence intervals; dammed basins show a flatter slope, the apparent buffering. (b) Differential slope (treat:SPEI; <0 = buffering) for intent-to-treat, downstream-only, and the upstream-only placebo; horizontal bars are 95% confidence intervals (estimate $\pm$ 1.96 cluster-robust SE), annotated with randomization-inference p-values. All are permutation-null, and the upstream placebo (above-dam, unregulated) is as strong as downstream, so the attenuation is siting/aridity, not the dam.

3.6. Storage declines without a loss of buffering capacity

If reservoirs neither buffer nor amplify drought operationally, the relevant change under the megadrought is in how much water is available to store (Figure 4). Over 2005–2024 the entire storage band drifts downward in step: annual peak storage declines by -1.3 percentage points of capacity per year ($p = 0.013$) and the annual trough falls almost identically (-1.2 pp yr⁻¹, $p < 10^{-4}$), while the seasonal amplitude is statistically unchanged (-0.06 pp yr⁻¹, $p = 0.91$). A band that drops without narrowing is consistent with a supply-side level shift rather than a degraded buffer, which would compress the amplitude; reservoirs still convert meteorological surplus into storage as efficiently as before. We caution that this storage trend is descriptive (a pooled within-reservoir trend with no control group or naturalized-inflow series), so it cannot by itself distinguish declining inflow from rising demand-side drawdown. The most direct demand proxy available to us weighs against the demand explanation, however: irrigated area in dammed basins shows no expansion over the period (stable, parallel trajectories, Figure 1; differential-expansion ATT indistinguishable from zero, Table 1), so a growing consumptive footprint is not evident as a driver of the storage decline, though a compositional shift toward more water-intensive perennials could raise per-area use without enlarging area. Read alongside Chile’s documented aridification, the pattern is therefore more consistent with a falling water supply than with any loss of storage or buffering capacity, and on that reading the operational case for expanding storage as drought protection is weak (Li et al., 2023; Wang et al., 2024).

4. Discussion

Across a matched national design and four independent estimators (cross-sectional irrigated-area expansion and ET buffering, and forcing-interacted slope gaps on observed area and ET), the operational effect of reservoirs on drought outcomes brackets zero once siting is matched out. The same answer holds for the direct, vegetation-independent availability outcome: the apparent flattening of meteorological-to-streamflow drought propagation in dammed basins is as strong at physically unregulated *upstream* reaches as downstream, and collapses from a slope gap of -0.18 to -0.03 when the comparison is made within climate region. Dammed basins do hold roughly 4.6 times more cropland than matched controls (11.1% vs 2.4% of basin area), but this gap is fixed across the record, with parallel pre-trends and no divergence through the megadrought. Taken together, the evidence supports a single interpretation: reservoirs **mark** where water-dependent vulnerability already concentrated (arid, already-irrigated basins), rather than **make** it. The vulnerability signature is a property of siting, set before the drought and stable through it, not a dynamic consequence of regulation.

The binding constraint is inflow, not storage

Whole storage band shifts down; seasonal amplitude unchanged (buffer intact)

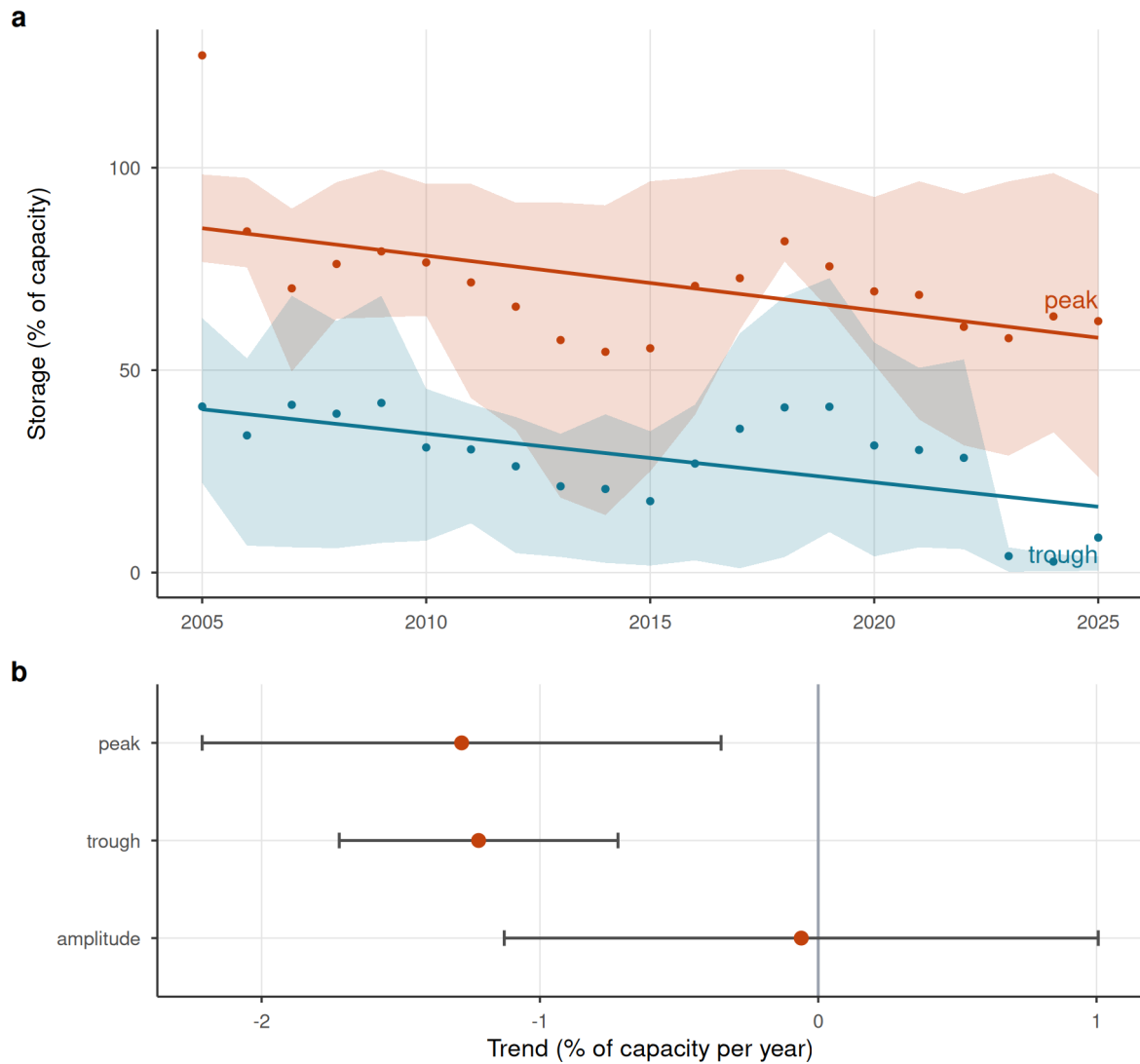


Figure 4: Storage declines without a loss of buffering capacity. (a) Cross-reservoir annual storage band, peak and trough percent-of-capacity (mean line; shaded bands are the interquartile range across reservoirs), drifts downward over 2005-2024 with roughly parallel trend lines: the whole band shifts while seasonal amplitude is unchanged. (b) Per-year trend slope of peak, trough, and amplitude (reservoir fixed effects); error bars are 95% confidence intervals (estimate $\pm$ 1.96 cluster-robust SE): peak and trough both decline while amplitude brackets zero, consistent with a supply-side level decline rather than a refill/buffer degradation. Descriptive: pooled within-reservoir trend, no control group.

This reframes “no effect” as the finding rather than a failure of power, but only because a credible *cause* of the apparent effect is identified. The within-basin upstream/downstream placebo is the cleanest within-design falsification here: a genuine regulation effect must appear downstream of the dam and be absent upstream, and it is not. What remains, once that placebo and within-region conditioning are applied, is the aridity gradient itself. The methodological corollary is general. A naive dammed–control contrast, or a calendar-time difference-in-differences, would read the large cross-sectional area, ET, and transmission gaps (amplified during the drought) as a reservoir effect; conditioning on realized meteorological forcing and estimating a deficit-to-impact *slope*, and benchmarking against an unregulated within-basin reach, makes the signal disappear. Reservoir-buffering studies that compare dammed against undammed basins, or pre- against post-construction levels, inherit the full siting confound (Lee et al., 2026; López-Moreno et al., 2009; Schilstra et al., 2024; Wu et al., 2017, 2018); the upstream placebo is a transferable identification tool applicable to any regulated river with gauges above and below the structure.

If reservoirs neither buffer nor amplify drought operationally, the relevant change under a multi-decadal dry spell is in the water available to store rather than in the structures, and the storage record is suggestive on this point. The seasonal refill remains efficient and its amplitude is unchanged, yet the entire storage band drifts downward, peak and trough declining in step at roughly -1.3 and -1.2 percentage points of capacity per year. A band that drops without narrowing is consistent with a falling supply ceiling rather than a degraded buffer. We stress that this is a descriptive, control-free trend that cannot on its own separate declining inflow from rising demand-side drawdown (Condon et al., 2020); both would shift the band downward without narrowing it. Our most direct demand proxy, however, weighs against the demand explanation: irrigated area in dammed basins did not expand over the period (stable, parallel trajectories; a differential-expansion ATT indistinguishable from zero), so rising consumptive demand is not evident as the driver, even though a crop-composition shift toward more water-intensive perennials could raise per-area use without enlarging area. Read against Chile’s documented transition from episodic drought toward structural aridification (in which declining precipitation and rising atmospheric evaporative demand lower inflows independently of any infrastructure (Alvarez-Garreton et al., 2021; Garreaud et al., 2017; Vicente-Serrano et al., 2022; Vicente-Serrano et al., 2019; Zambrano et al., 2025), echoing aridification trajectories documented elsewhere (Overpeck and Udall, 2020)) and the global pattern in which reservoirs increasingly fail to refill and the storage returns of new dam construction are diminishing (Li et al., 2023; Wang et al., 2024), the more likely reading is that the binding constraint is shifting toward water supply rather than storage capacity or operation. A naturalized-inflow series and a demand-side decomposition are needed to settle this, and we

treat it as a strong hypothesis rather than an identified result.

The policy implication is direct and uncomfortable for the prevailing response to scarcity in Chile and elsewhere. Because the 1981 Water Code makes existing water-use rights perpetual private property that the state cannot readily curtail during drought, the reflexive response has been supply-side, a national programme of new dams and inter-basin transfers (Barría et al., 2021; Garreaud et al., 2025; Macpherson et al., 2023). Our results imply that adding storage to basins whose inflow is declining is a supply-blind response to a supply problem: it cannot manufacture water that aridification is removing, and the buffering benefit it is built to provide is, on this evidence, attributable to where dams sit rather than to what they do. This is the drought-domain expression of the reservoir-effect and supply-demand-cycle hypotheses (Di Baldassarre et al., 2018), whose practical force has been contested for want of quantification (Kellner, 2021); we provide a credibly identified test and find the protective function over-attributed. The leverage instead lies on the demand and allocation side (curbing the over-allocation and induced expansion that the efficiency paradox entrenches (Grafton et al., 2018; Jordán et al., 2025; Reade Malagueño and D’Odorico, 2024)) and in managed recharge and reallocation rather than new impoundments (Rivera et al., 2016; Rivera-Vidal et al., 2025; Scanlon et al., 2023).

Several limitations qualify these conclusions. First, treatment is effectively time-invariant over the record (most reservoirs predate it), so we identify differential drought *response* rather than the effect of switching a reservoir on; a staggered-commissioning design is infeasible here and would be the cleaner counterfactual where construction dates and longer records permit. Second, the result is a bounded null, not a proven equivalence: with ~21 treated clusters the design rules out buffering larger than 46–58% of baseline transmission for streamflow but cannot exclude small operational effects, so the causal weight rests on the placebo, not on equivalence. Third, the observed-area outcome (MapBiomass class 18) captures total cropland rather than irrigation status, and Chile’s agricultural change has been compositional (a shift toward perennial orchards within roughly constant cropland area), so a latent water-demand signal not visible in net area remains possible; whole-basin MODIS ET is additionally confounded by residual aridity over barren land, so the ET conclusion leans on the cross-sectional buffering null and the orchard stratum, both low-powered (minimum detectable effects of roughly twice their baselines), so for ET specifically the finding is “no *large* effect detected” rather than no effect. Relatedly, the matched set leaves baseline aridity imbalanced (SMD 0.17), so the doubly-robust adjustment for it is load-bearing and the estimates are sensitive to its functional form: a quadratic aridity term moves the proxy transmission-slope estimate from +0.33 to +0.11, toward

the null we report, and a reminder that the linear-adjustment values are not the conservative ones. Fourth, the streamflow placebo covers 15 of 21 treated basins over 2000–2020, and the high-supply southern subset is small; a modest regulation effect in wetter basins cannot be excluded, though drought is also least binding there. Upstream and downstream gauges are also not perfectly interchangeable (upstream gauges sit higher, and elevation predicts the transmission slope among controls), so a residual regime component remains; that said, the raw upstream and downstream transmission slopes are nearly identical and the apparent buffering is if anything stronger upstream, neither of which is consistent with a regulation effect obscured by elevation. The 12-month aggregation is a further boundary: it targets annual-to-interannual transmission and does not resolve the sub-seasonal (1–3 month) operational buffering reservoirs are designed to provide, so our conclusions concern long-term vulnerability rather than short-term operation. Generalizability is also bounded by governance: Chile’s 1981 Water Code is an unusually extreme, market-based regime (perpetual, tradable private rights the state cannot readily curtail), so while the *operational* finding (that buffering reflects siting, not regulation) should hold wherever dam siting is non-random, the demand-feedback and policy inferences are conditioned on this weak adaptive feedback; under state-managed or dynamically regulated allocation, operators able to curtail demand or re-time releases could in principle produce genuine buffering, or weaker induced-demand lock-in, than we observe here. Finally, residual SUTVA spillovers (adjacency, shared aquifers) survive the contamination rule, and our estimand is the *drought* effect of reservoirs, not their value for hydropower, flood control, or urban supply, which this study does not assess.

One alternative interpretation deserves emphasis because mean-based estimators cannot see it: damming might leave the *magnitude* of drought impact unchanged while recomposing its **dependence structure**, weakening the meteorology→outcome coupling and substituting a storage→outcome coupling, a regime shift that would itself explain the convergent null. We cannot adjudicate this with the present data: the storage-recovery dynamics are stationary (refill efficiency and amplitude are flat), which weakens the premise, and the decisive test requires reservoir release and operating-rule records that set storage *independently* of contemporaneous inflow. The only storage signal on disk is monthly change in storage (a drawdown proxy), which is an endogenous hydrological response to the same precipitation and inflow forcing (and partly to demand) and is therefore collinear with SPEI; regressing an outcome on it cannot separate a genuine storage→outcome regime from the forcing→outcome coupling we already estimate, because it carries no independent allocation information. Identifying the dependence-shift limb thus needs exogenous release or rule-curve data. Absent that, and given that any apparent decoupling must survive the upstream placebo, the dependence-shift account reduces to the siting interpretation. Resolving it (together with longer streamflow

records and water-rights allocation data) is the priority for future work.

Returning to the question that motivates the study (do reservoirs create resilience or merely postpone vulnerability under prolonged drought?), our evidence indicates that, under aridification, they do neither operationally: storage neither buffers nor amplifies the drought signal beyond what basin siting already implies, while the water available to store steadily declines. Resilience to a falling supply ceiling will not come from more impoundments but from confronting demand and allocation, the harder, and unavoidable, adaptation.

5. Data and code availability

The full analysis pipeline, comprising the `targets` workflow, the reusable R functions (`src/R/`), the study-parameter configuration (`config/`), and the scripts that generate every figure and table, is openly available at https://github.com/frzambra/dam_drought_effectiveness and archived on Zenodo (DOI: 10.5281/zenodo.XXXXXXX). The pinned R environment (`renv.lock`) reproduces the package versions used. Reservoir storage records, the derived result tables, and matched-set metadata are included in the repository; the third-party input datasets are publicly available from their providers as cited: CHIRPS/CHIRTS (standardized drought indices), MODIS MOD16/MOD13/MOD17 (evapotranspiration and vegetation), MapBiomass Chile (land cover), the CR2 network (streamflow), SRTM (elevation), and the DGA reservoir and watershed registries. Every result in the manuscript corresponds to a named pipeline target, so the analyses can be regenerated with a single `targets::tar_make()`.

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